

SRINITHI K

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PROFESSIONAL SUMMARY

AI systems engineer specializing in production-grade, cloud-native RAG and multi-agent architectures for real-world industrial applications, with an interest in integrating AI with embedded and hardware-driven environments.

PROFESSIONAL EXPERIENCE

GGG Information Services, AI Engineer

2025 Jul – Present
Chennai, India

- Architected and documented the end-to-end design of a **production-grade, RAG-based agentic AI platform** for the automotive OEM aftermarket domain and took developmental ownership of 3 features.
- Led and mentored junior engineers and handled client interaction. Driven the meeting with Google team to validate the architecture and the use of GCP services in it.
- Designed and implemented a **multi-agent conversational system using LangGraph**, orchestrating workflows across structured (tables) and unstructured (text document, videos) enterprise datasets.
- Built and deployed a **custom ingestion + hybrid retrieval pipeline (Dense Embeddings + BM25)** of service procedure (XML) on Google Cloud Vertex AI and Cloud Run, enabling real-time contextual assistance for service technicians and reducing service downtime.
- Developed the microservices in **Python** using FastAPI and deployed it as a **cloud-native microservice on GCP**, ensuring modular architecture, observability, and enterprise reliability.
- Integrated **LLM safety and policy enforcement using Model Armor, sensitive data protection policy in GCP**
- Architected the design in Google ADK for comparison built a POC for **voice-enabled documentation (using Whisper) , LangGraph based flowchart assistant and AI-driven content automation pipelines** to streamline manual creation and improve operational scalability.

Tech Stack: GCP, Foundational Model, Embedding model, BM25, Hybrid Search, Whisper, Ollama, Microservice architecture, RAG, Agentic AI, LangGraph, LangSmith, Vertex AI Vector Store, Postgres DB, Cloud SQL, Google ADK, NLTK, Pydantic, LlamaGuard, Model Armor, Prompt engineering, XML, JSON, SVG, GIT, RAGAS.

AMAZON, Software Development Engineer Intern

2025 Jan – 2025 Jun
Chennai, India

- Contributed to the **Search Application** on Amazon Tablet devices, capturing search performance metrics and cross-language consistency.
- **Analyzed metrics** and defined detailed logging requirements to improve observability and user behavior tracking.
- **Redesigned and rearchitected** the metrics logging system unifying metrics collection across Kotlin (Android) and React Native components using builder and singleton design pattern, enabling accurate and consistent performance tracking from both the packages.
- **Collaborated** with multiple teams and stakeholders to build and validate the unified telemetry pipeline, ensuring **long-term maintainability and reliable analytics** across product teams.

Tech Stack: AWS, Java, Kotlin, React Native, GIT

Robotico Digital LLP, AI Intern

2024 Jun – 2024 Jul
Chennai, India

- Created a custom neural network model with **LSTM** to predict the future work order with 10% loss value.
- Worked with LSTM, **weibull-RNN models for time series prediction**.

RBG.AI, AI Intern

2024 Jan – 2024 May
Chennai, India

- Developed a web application that responds to the question by finding the answers from the text content of the uploaded image using **Large Language Models** with 68% accuracy.
- Conducted a comparative analysis of **YOLOv8–v10** for wildlife detection in camera trap images (16K+ images, 7 species). Achieved **95.6% precision** with YOLOv10, addressing challenges like lighting, pose, and background clutter.
- Annotated dataset using Roboflow, evaluated models on mAP, precision, and recall. **Published a paper in IEEE "Wildlife Recognition in Trap Images: Assessing the Efficiency of YOLO Frameworks"** (paper id - 10939902)

TECHNICAL SKILLS

Languages: Python, Java, C, SQL (PostgreSQL, MySQL), Kotlin, JavaScript, R, React Native, HTML, CSS

AI / Machine Learning: Retrieval-Augmented Generation (RAG), Agentic AI, Hybrid Retrieval (Dense Embeddings + BM25 + TF-IDF), LLM Safety (LlamaGuard), Prompt Engineering, NLP (NLTK), YOLO (v7–v10), LSTM, Time Series Modeling, Exploratory Data Analysis, Image Preprocessing (Gaussian Blur)

Cloud & Backend Engineering: Google Cloud (Vertex AI, Cloud SQL), Firebase (Realtime DB, Firestore), Microservices Architecture, REST APIs, Flask.

AI Frameworks & Tools: LangGraph, LangSmith, Google ADK, Whisper, Ollama, NLTK, Pydantic, SpeechRecognizer, Pyttsx3, LXML, Streamlit, Roboflow

Databases: PostgreSQL, Firebase, Mango DB

Developer Tooling: Git, Power BI, R Studio, Roboflow, Android Studio, VS Code, Google Colab, MATLAB, Tinkercad, MIT APP Inventor, Canva, Cursor

PROJECTS

Eco Vision

- Preprocessed animal trap images from a conservation dataset images with Gaussian blur and resized the images. Wrote a code snippet to obtain a balanced dataset of 200 images from each class.
- Trained Yolo v7, v8 and v9 models on the dataset and conducted a comparative study. Performed hyperparameter tuning to improve the models.
- Annotated the dataset manually in Roboflow with smart polygon bounding box.

Convo Buddy

Paper titled "E-Companion: Revolutionizing Patient Mental Care", Accepted for publication in API Conference Publications.

- Implemented a Retrieval Augmented Generation (RAG) voice bot, trained on a custom dataset to handle basic conversational queries with a bedridden patients. Developed an application to send personalized data via Firebase to respond with customized answers along with realtime health data from sensors.

Igniters, TN EDII Finalist project

- A realtime mobile and web application is built using java to control the street light efficiently and effectively for sustainable usage with accurate fault detection of light, using a cloud Firebase as its database.
- Various sensors such as ldr sensor, rain sensor, gas sensor, humidity sensor are used to detect environmental conditions and using ESP8266 and a network of controllers are used to automate street lighting.

Tech Stack: Java, Android Studio, HTML, CSS, JS, Arduino IDE, Google Cloud - Firebase (RealtimeDB, Firestore, Authentication), Gas sensor, LDR, Humidity sensor, MUX, Solar cell, ESP8266, Arduino.

EDUCATION

B.E. Computer Science and Engineering , St. Joseph's College of Engineering Chennai, India	2021 – 2025 CGPA 8.98/10.0
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AWARDS

Finalist , Entrepreneurship Development Institute of India - Tamil Nadu	2024 Sep
Finalist , AnalyticaX - CFA Conclave'24, IIT Indore	2024 Mar

CERTIFICATES

BEC VANTAGE LEVEL B2 CAMBRIDGE UNIVERSITY - Business English Communication	Python For Data Science (ELITE-Silver) NPTEL - IIT Madras	Introduction to Probability and data with R COURSERA - Duke University
Google Cloud Computing Foundation (ELITE - Silver) NPTEL - IIT Karagpur (Top 5%)	Computer Vision Onramp MATLAB - Mathworks	Machine learning and its applications using Python EduxLabs with Mechanica IIT Madras

CODING PERFORMANCE

LeetCode

- Solved **150+ coding problems**.
- Earned badges: **100 Days Badge (2024)**, **Introduction to Pandas**, **January LeetCode Challenge**.
- Profile: <https://leetcode.com/u/Sri15nithi/> 

VOLUNTEER

AI PEP, Embeddings'24 Acted as a member of the AI community at college and organized the event Embeddings'24.	CSS BATTLE, CRYPTIX'23 Organized an event in CSE department's Symposium.
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